

TITLE: Model Deployment using FastAPI

1. Objective

The objective of this task is to deploy the trained machine learning model as a web API using FastAPI.

2. Model Used

Random Forest Classifier trained on the Titanic dataset.

3. Model Saving

The trained model was saved using Joblib.

File:

titanic_model.pkl

4. API Development

FastAPI was used to create API endpoints.

Endpoints:

GET /

Returns API status message

POST /predict

Returns prediction result

5. Deployment Process

- Saved trained model
- Created app.py file
- Loaded model using Joblib

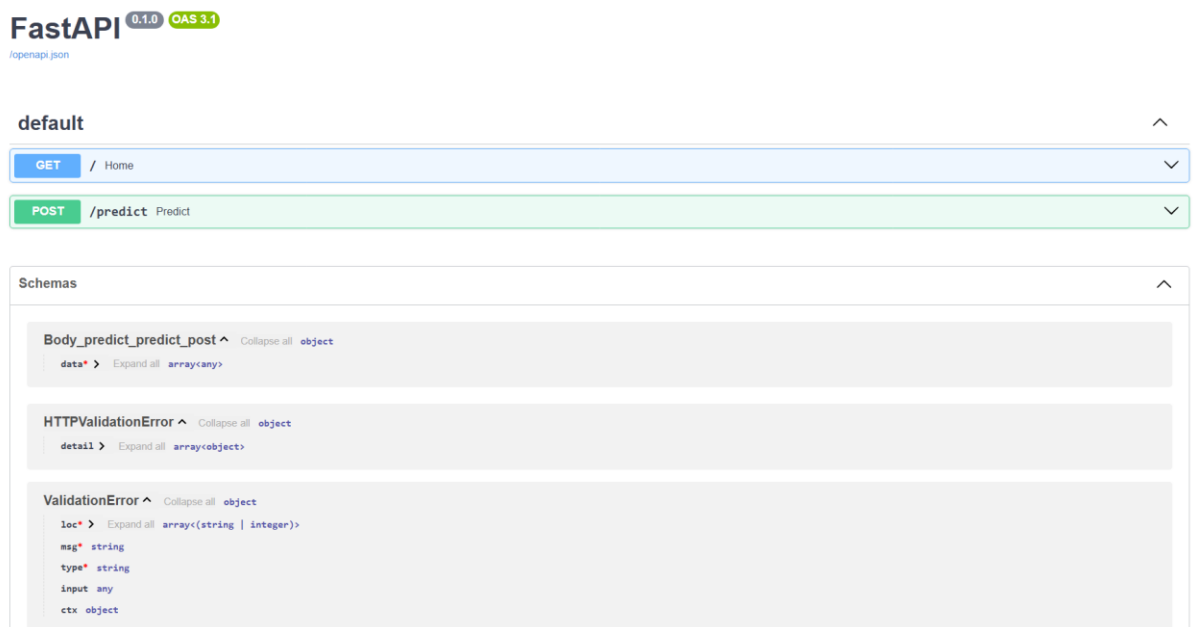
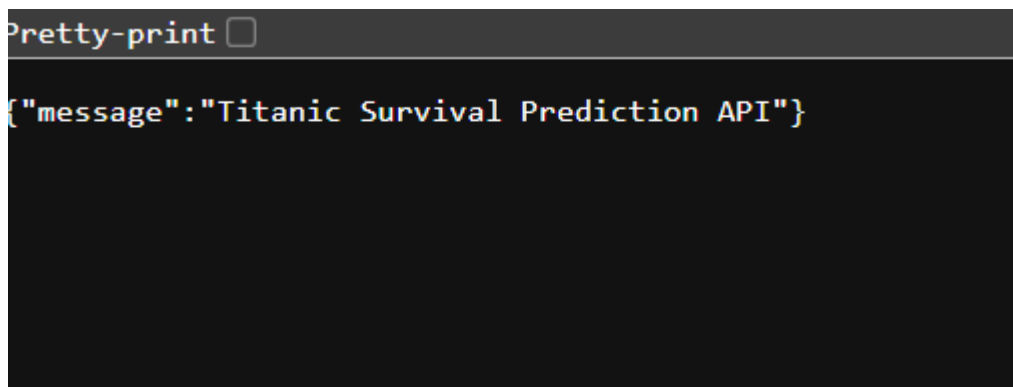
- Started FastAPI server using Uvicorn
- Tested API successfully

6. Results

The API successfully responded with:

```
{"message": "Titanic Survival Prediction API"}
```

7. Screenshots



8. Conclusion

The machine learning model was successfully deployed as a FastAPI application and can be accessed through API endpoints for prediction.